AI6127 Assignment - Editable Working Copy

> **Repository-backed draft, updated 2026-11-03 (Singapore).** Assignment prompts remain separate from the added answers below. Existing user edits to the assignment text are preserved. “Implemented” refers to inspected code; historical/local observations are attributed to their recorded source and are not fresh measurements. Final evaluation remains pending while two labeling files are unfinished. This is a working report draft, not a completed submission.

Evidence snapshot	Revision inspected	Scope
`origin/main`	`77626799cee68b27059760c2c20e044f61cebee2`	Maintained hybrid, agent and school generation pipeline
`origin/p3-qasper-foundation` (unmerged)	`c612ef362e8dcbfed62103799140e9a34df17f3d`	Current corpus, chunk-size experiments and annotation pool
UI PR #20 (unmerged per user)	`18fe2b3d3f02a631491c9b6f8bfbdee40f668802`	Question-first Streamlit UI, local Ollama integration and corpus search

Remote refs and PR head were fetched before inspection. Evidence links below pin exact commits. Older planning documents can lag implementations; this draft uses the newer artifact manifests for corpus/pool facts and preserves `v0-proposed` status for shared passage schemas and split mappings. A source path/commit is not proof that a historical run used a clean checkout: several manifests/docs explicitly record dirty source hashes. Raw runtime artifacts named in local guides were unavailable here, so those run observations remain source-reported.

How to read status and ownership

- **Completed** means the specific answer is documented and supported at the pinned scope; it does not imply all related branches are merged or the project is complete.
- **In Progress** uses an editorial five-star estimate: ★☆☆☆☆ = setup/protocol; ★★☆☆☆ = partial evidence; ★★★☆☆ = implementation/partial runs; ★★★★☆ = mostly implemented or labeled, with required verification outstanding. Stars are not measured accuracy, a rubric grade or an exact completion percentage.
- **Not started** means the required execution has not begun or is not evidenced; planning notes may already exist.
- Owners below are responsible **roles inferred from repository module ownership**, not newly assigned people. Named assignees and human approval remain pending confirmation. Updated dates refer to this workbook edit, not a new experiment.
- Bullets inside each summary table use HTML lists for Markdown table compatibility. Detailed results, qualifications and flowcharts are outside the summary table so they remain editable.

1 Objective

In this group assignment, you will build a retrieval-augmented generation (RAG) system for a natural language processing task of your choice. You are free to pick any downstream task that retrieved evidence can serve, e.g., open-domain question answering, summarization, fact-checking, semantic search, recommendation, or sentiment and opinion analysis (for instance, a system that finds relevant opinions about any instance of a topic of your choice, such as bitcoin within the topic of cryptocurrencies, and reports that they are 70% positive and 30% negative). Whatever task you choose, the heart of your system must be the retriever-generator contract of this course: a knowledge base that is gathered and indexed offline, a retriever that locates the most relevant evidence for a query, and a generator (or classifier) that produces the final answer conditioned on that evidence. Once you have chosen a task and a domain, make sure that a) you can gather enough data about it (e.g., some topics may be too niche to the point that you would only find a few hundreds documents about it) and b) the data are suitable and balanced for the task (e.g., for sentiment analysis, if the topic you chose only has negative opinions associated with it, it's probably not a good topic). For ideas about interesting topics, you can check our project page at https://sentic.net/projects. You are pretty much free to use anything you want in terms of available tools/libraries. However, your system cannot be just a mashup of existing services or a single call to a hosted RAG API: you must implement, and be able to explain, the retrieval and generation stages yourselves. Your final score will depend not only on how you developed your system but also on its novelty and your creativity: in other words, to get a high score you do not only need to implement a system that works, but also a system that is useful and user-friendly.

2 Deadline and Grouping

- The assignment constitutes 35% of your total grade for the course.
- Report submitted by 7th November at 11:59pm SGT.
- The main tasks of the assignment are knowledge-base construction (20 points), retrieval (40 points) and the downstream task with generation (40 points).

3 Tasks and Questions

3.1 Knowledge-Base Construction (20 points)

Assemble the corpus that your system will retrieve from. Gather text data from any sources which you are interested in and permitted to access, e.g., Reddit API or a public dataset, and/or combine them with ready-made datasets. The corpus should have at least 10,000 documents (or passages) and at least 100,000 words. It is OK to use available datasets for training or as background knowledge (e.g., popular sentiment benchmarks or datasets from the Hugging Face Hub), but you still have to at least gather and label a held-out test set yourselves. For your own evaluation dataset, make sure it does not contain duplicates and try your best to make it balanced (e.g., an equal number of positive and negative entries for sentiment). For a classification task, use the same tabular format as the above-mentioned sentiment benchmarks and name it "eval.xls" (using a different format will result in demerit points); for other tasks, use an appropriate standard format (e.g., JSONL) and document it. Because a generator will read what you retrieve, you must also clean and chunk the raw documents into passages of a sensible size, which is itself a design choice discussed in the course. You can use any third-party libraries for gathering and preparing the data, e.g.:

- Hugging Face Datasets: https://huggingface.co/datasets

- Kaggle Datasets: https://www.kaggle.com/datasets
- Common Crawl: https://commoncrawl.org
- Wikimedia dumps: https://dumps.wikimedia.org
- Scrapy: https://scrapy.org
- Beautiful Soup: https://www.crummy.com/software/BeautifulSoup
- Trafilatura: https://trafilatura.readthedocs.io
- Playwright: https://playwright.dev
- PRAW (Reddit): https://praw.readthedocs.io

Note that access to social-media data has tightened in recent years: the X (formerly Twitter) API now requires a paid tier and several platforms have restricted their public APIs, so prefer open datasets and permissively licensed sources, and always respect each source's terms of use and rate limits.

Question 1

Explain and provide the following:

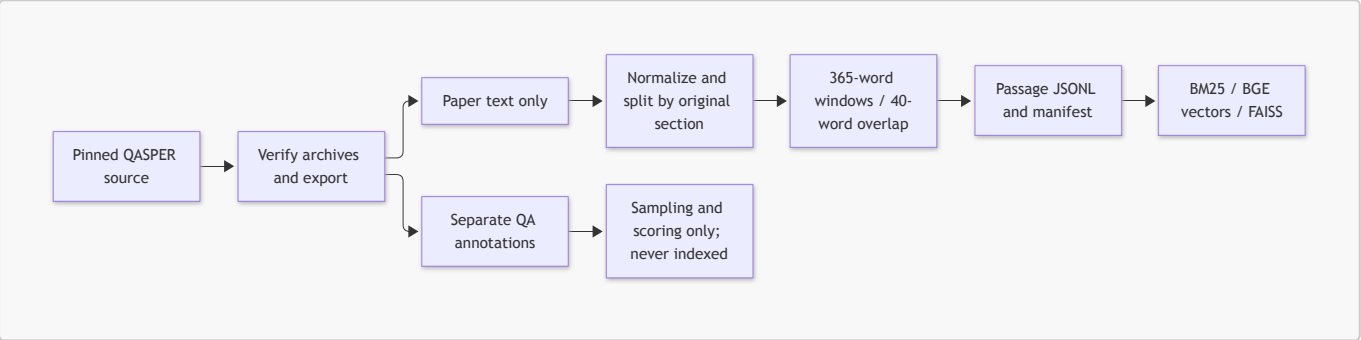
Q1.1

How you gathered the corpus (e.g., source, keywords, API, library), and how you cleaned, chunked, and stored it

Field	Details
Status:	• Completed • This scoped response is supported by the inspected evidence; not a claim of project-wide completion.
Answer:	• Pinned QASPER acquisition; separate paper/QA exports; text-only indexing. • Conservative cleaning; section-contained 365-word chunks with 40-word overlap; JSONL plus hashes/manifests. • See supporting details and any standalone tables below.
Decision / rationale:	• Use the current P3 365/40 corpus description; keep older 200/40 experiments separate. • Passage schema remains `v0-proposed`; changing corpus requires matching indexes and evidence mappings.
Evidence / links:	• Acquisition implementation • Chunker • P3 reproduction guide

Field	Details
	• Passage schema
Owner / updated:	• Responsible roles: P3; P1/P2 for index compatibility. • Named assignees/approval: pending confirmation. • Workbook entry updated: 2026-11-03; underlying evidence remains at the pinned snapshot.

Flow / workflow



Answer details

- PaperGap uses QASPER scientific-paper text (`allenai/qasper`) as its knowledge base. Acquisition is pinned to dataset revision `fdc9d8214fbab5dd782958601db4d678e6934a54`, source version `0.3.0`. The downloader checks source metadata, archive fingerprints and required members and parses the source locally; it does not execute the Hub dataset script. This is public-dataset acquisition, not keyword scraping or social-media collection.

- The export separates `papers.jsonl` (paper text/metadata) from `qa.jsonl` (questions and original answer annotations), retaining official split provenance. Only `id`, `title` and `full_text` enter passage construction; answers, gold evidence and questions are not indexed. Cleaning normalizes CRLF/CR to LF, trims outer whitespace, and removes unusable empty containers. It preserves case, punctuation, formulas, internal spacing and citation placeholders rather than stemming or rewriting scientific text.

- The current published P3 corpus uses **365 words per chunk and 40 words of overlap**, with Unicode non-whitespace runs as words. Windows stay inside one original full-text section; paragraphs within that section may be joined. Repeated section names do not merge source containers. Abstracts and figure/table sources are not indexed by this builder. Passage IDs are deterministic opaque `qasper-passage-v1-…` hashes of source/configuration coordinates. The eight public fields are `passage_id`, `paper_id`, `title`, `section`, `chunk_number`, `text`, `prev_passage_id` and `next_passage_id`; neighbor links stay inside the same container. Schema status remains `v0-proposed`.

- Passages are stored as JSONL with companion manifests recording hashes, source revision, normalization/chunking settings and corpus counts. Validation checks unique IDs, nonempty text and valid mutual neighbor links. BM25 consumes passage text; dense retrieval uses saved BGE

vectors and FAISS. Reproduction follows the P3 `data/README.md` and its download/export/build/check commands.

Decision / rationale details

- **Version decision:** use the 365/40 manifests for the current KB description. Older P1/P2 integration results used 200/40 chunks and 32,073 passages; they are historical results, not measurements of this corpus. `configs/default.yaml` on main is explicitly an unconsumed proposal and still says 200, so it is not evidence of the newer build configuration. A changed corpus needs matching indexes and evidence mappings. Raw data, generated corpora, weights and indexes must not be newly committed as part of this documentation update.

Q1.2

Which task users will perform over your corpus (i.e., the application), with sample queries or inputs

Field	Details
Status:	• Completed • This scoped response is supported by the inspected evidence; not a claim of project-wide completion.
Answer:	• Scientific QA about methods, architectures, experiments, results and comparisons. • Answer from cited passages or refuse; paper-scoped main API and PR-only corpus search are distinct. • See supporting details and any standalone tables below.
Decision / rationale:	• Clarify paper scope for ambiguous questions; use existing development examples. • Describe the formal annotation comparison as paper-scoped, not corpus-wide.
Evidence / links:	• Development pilot inputs • Main hybrid API • PR corpus retriever • UI scope and example controls
Owner / updated:	• Responsible roles: P3/P5; P6 for application scope. • Named assignees/approval: pending confirmation. • Workbook entry updated: 2026-11-03; underlying evidence remains at the pinned snapshot.

Answer details

- Users ask questions about NLP research papers: methods, model architecture, experiment settings, results and comparisons. The system retrieves passages, optionally diagnoses evidence gaps and retries, then generates an answer with passage citations and provenance, or refuses when evidence is inadequate. This is a generative QA application, not sentiment classification.
- Main's maintained hybrid API retrieves **within a selected paper**. PR #20 adds corpus-wide retrieval and an optional one-paper filter; its prepared local corpus contains the 1,168 indexed development papers. Corpus search is not automatically enabled by the main paper-scoped interface. The UI exposes it only when `/health` advertises `corpus_search`. The formal P3 four-method annotation pool evaluates selected-paper retrieval, not corpus-wide search.

Examples from the already-used five-question development pilot:

- Paper `1912.10167`: “Are any experiments performed to try this approach to word embeddings?”
- Paper `1911.07555`: “What evaluation metric is used?”
- Paper `1904.10503`: “What results do they achieve using their proposed approach?”
- Paper `1904.05584`: “Which model architecture do they use to obtain representations?”
- Paper `1912.01046`: “What evaluation metrics were used in the experiment?”
- A corpus-level UI example is “What methods are used for low-resource language identification?” Questions containing “they” or “this approach” need the selected paper or an explicit paper reference. These examples are development inputs, not newly sampled final-evaluation questions.

Q1.3

The numbers of documents (and chunks), words, and types (i.e., unique words) in the corpus

Field	Details
Status:	• Completed • This scoped response is supported by the inspected evidence; not a claim of project-wide completion.
Answer:	• Development: 21,360 passages; 4,262,446 source words; 174,636 casefolded passage-word types. • All-paper-text inference: 28,617 passages; 5,659,921 source words; 210,909 types. See corpus table below. • See supporting details and any standalone tables below.
Decision / rationale:	• Report source papers, indexed papers, source words and overlap-inflated passage words separately. • Treat counts as verified manifest snapshots; distinguish the full KB from the evaluation index subset.

Field	Details
Evidence / links:	• Development manifest • Inference manifest • Formal pool manifest
Owner / updated:	• Responsible roles: P3. • Named assignees/approval: pending confirmation. • Workbook entry updated: 2026-11-03; underlying evidence remains at the pinned snapshot.

Answer details

Measure	Development corpus	All-paper-text inference corpus
Source papers	1,169	1,585
Papers with indexed passages	1,168	1,584
Unique passages / indexed documents	21,360	28,617
Source words, before overlapping windows	4,262,446	5,659,921
Passage words, including overlap	4,469,326	5,930,961
Casefolded passage-word types	174,636	210,909

- Both corpora exceed 10,000 passages and 100,000 source words. The team recorded that passages count as documents for the rubric. Source papers and passages are reported separately; overlapping windows must not inflate the source-word total. Types here mean distinct casefolded non-whitespace runs in emitted passage text; punctuation remains part of this operational definition, so these are not linguistically normalized vocabulary counts. One source paper contributes no usable indexed full-text passages.
- The development corpus contains official train + validation paper text (16,457 + 4,903 passages). The inference corpus additionally includes 7,257 passages from official test papers. The export contains 5,049 questions and 7,993 answer annotations across 1,585 papers; those QA annotations are not corpus documents.

Corpus SHA-256 values:

- Development: `ecb1c601cdaebcd8e1e6bfeaf43f9cb4867f89e9762799ab06e7901549a9e403`.
- All paper text: `8eeef0119e322fc9253255e2c6db6cb38ec276d7ebf3669dd28a856a636aa93e`.

Decision / rationale details

- **Scope decision:** the prepared local UI uses the development corpus. The formal annotation manifest records a retrieval index of 3,169 passages from the 174 selected papers, drawn from the

all-paper-text corpus; that evaluation subset is not the complete KB. These counts are manifest-backed snapshots, not a fresh recount of raw data during this edit.

3.2 Retrieval (40 points)

Retrieval: Build a retriever over your knowledge base. At a minimum, implement both a sparse (lexical) retriever and a dense (embedding) retriever, and combine them into a hybrid pipeline, optionally followed by a re-ranker, exactly the design developed in this course. You can build each stage from scratch or use some combination of available tools. Useful components include:

- Elasticsearch / OpenSearch (inverted index, BM25): https://www.elastic.co, https://opensearch.org
- Apache Solr / Lucene: https://solr.apache.org, https://lucene.apache.org
- Pyserini (reproducible BM25 and dense retrieval): https://github.com/castorini/pyserini
- Sentence-Transformers (SBERT embeddings): https://sbert.net
- Hugging Face Transformers (BGE, GTE, E5 encoders): https://huggingface.co/models
- FAISS (similarity search library): https://github.com/facebookresearch/faiss
- Vector databases: Chroma, Qdrant, Weaviate, Milvus, pgvector
- RAG orchestration frameworks: LangChain, LlamaIndex, Haystack, DSPy

You can also choose any other inverted-index or vector-search open project. However, you should NOT simply adopt SQL-based solutions for text search (for example, you CANNOT solve retrieval simply using a LIKE query in Microsoft SQL Server or MySQL).

Querying: You need to provide a simple but friendly user interface (UI) for querying. It could be either a web-based or mobile app based UI. Lightweight options such as Streamlit or Gradio let you build one in a few lines; FastAPI or Django are options for a fuller web app. The UI must be kept simple. A sophisticated UI is not necessary nor encouraged (as it is not the focus of this course). Detailed information besides text is allowed to be shown for the query results, e.g., images, ratings, timestamps, and source links. The details should be designed to solve specific problems.

Question 2

Perform the following tasks:

Q2.1

Design a simple UI (you can design one from scratch or you can tap on an existing one) to allow users to query your system in a simple way

Field	Details
Status:	• In Progress — ★★★★☆ (4/5)

Field	Details
Answer:	• Streamlit question input, scope selection, example library, history, citations, warnings and run export are implemented in PR #20. • Local real-model integration is documented; school GPU rehearsal and merge/review remain pending. • See supporting details and any standalone tables below.
Decision / rationale:	• Use backend capabilities to enable corpus search and paper filtering. • Keep prepared-record generation distinct from fresh retrieval; keep local Ollama results distinct from school Qwen3-8B/vLLM.
Evidence / links:	• UI implementation • HTTP client • Completion evidence and limitations • Local real-environment guide
Owner / updated:	• Responsible roles: P6; P5 for HTTP service; P4/P1/P2 for pipeline. • Named assignees/approval: pending confirmation. • Workbook entry updated: 2026-11-03; underlying evidence remains at the pinned snapshot.

Answer details

- The Streamlit application is titled **NLP Scientific Paper RAG**. The sidebar contains connection status, search scope, history and advanced settings. The main area starts with an editable question, an **Ask question** button and an expandable example library, followed by the submitted question and one selected answer. Paper-specific examples display titles and IDs; selecting an example fills the draft without submitting it.

- Users choose **All papers** when the backend advertises corpus search, or **One paper** with a title/ID filter. Each question is independent: session history is not sent as conversational context. Answers show evidence status, warnings or validator blocks, expandable sources with passage text, pipeline details and debug metadata. A refusal is a valid answer outcome, distinct from an HTTP, parsing or generation failure. Prepared-record mode uses supplied evidence and generates a new answer; it does not establish fresh retrieval or fresh agent execution.

- The UI talks to a P5-compatible HTTP service, using `/health`, `/papers`, `/answer` and `/passages`. It does not call the LLM directly. History is session-only; sanitized JSON export can capture five selected runs. Timeouts leave the service outcome unknown; retry is manual. No streaming or cancellation is promised.

- The PR's local integration uses Ollama `qwen3:14b` plus real development text, BM25/BGE/RRF/reranking, P4's adaptive agent and P5 validation. School deployment is separately designed around Qwen3-8B/vLLM. The PR documents local real-model and browser checks; the

latest layout regression record is 98 passed and one optional running-service test skipped. Earlier larger/subset suites are different runs, not additive totals. School GPU rehearsal remains pending.

Decision / rationale details

- **Decision:** describe this as PR functionality with documented local integration, not as merged-main functionality or a completed school deployment.

Q2.2

Write five queries, get their results, and measure the speed of the querying

Field	Details
Status:	• In Progress — ★★☆☆☆ (2/5)
Answer:	• Five existing development queries are listed below. • Separate local integration timings exist, but five matched answer/retrieval/latency exports are missing. • See supporting details and any standalone tables below.
Decision / rationale:	• Run/export the same five inputs under one fixed configuration; retain refusals and warnings. • Measure answer HTTP and citation lookup separately; record full API traces and provenance.
Evidence / links:	• Five development inputs • Observed local executions and timing boundaries • Run export implementation
Owner / updated:	• Responsible roles: P6; P5/P4 for execution support. • Named assignees/approval: pending confirmation. • Workbook entry updated: 2026-11-03; underlying evidence remains at the pinned snapshot.

Answer details

- Use the existing development pilot below, keeping paper scope explicit. The repository evidence reviewed does not supply a complete matched table of answers, retrieved IDs and per-query live latency for all five. It would be inaccurate to invent those outputs or assign aggregate timings to individual questions.

#	Paper ID	Query	Answer and retrieved passage IDs	Measured query latency
1	1912.10167	Are any experiments performed to try this approach to word embeddings?	Pending saved live-run export	Pending
2	1911.07555	What evaluation metric is used?	Pending saved live-run export	Pending
3	1904.10503	What results do they achieve using their proposed approach?	Pending saved live-run export	Pending
4	1904.05584	Which model architecture do they use to obtain representations?	Pending saved live-run export	Pending
5	1912.01046	What evaluation metrics were used in the experiment?	Pending saved live-run export	Pending

- The PR documents real local observations on Apple M5 / 24 GB using Ollama Qwen3-14B: one paper-filtered browser request took **77,779 ms** over three rounds and retained insufficient-evidence/component warnings; one corpus-mode AppTest took **79.42 s**, cited two of four retrieved papers and retained `UNSUPPORTED_CLAIM`; a later corpus-mode browser request took **75.16 s**, searched three papers over three rounds and returned a refusal with no citations. These are separate integration observations, not five independent benchmark rows, and their scientific correctness was not established. The raw ignored runtime files named by the guide were not available for independent reinspection in this checkout.

Decision / rationale details

- **To finish this answer:** run each of the five inputs in the same fixed environment, export those five runs, and save the direct API responses for full agent traces. Record corpus/index hashes, source identity, generator model/digest, prompt, scope, caching/warm-up policy and timestamps. Report the answer HTTP duration separately from citation lookup and from retrieval/model-only timing. Preserve refusals and warnings as observed. Do not call UI elapsed time “inference latency,” use synthetic mock timing, or reuse the held-out annotation questions for demo tuning.

Question 3

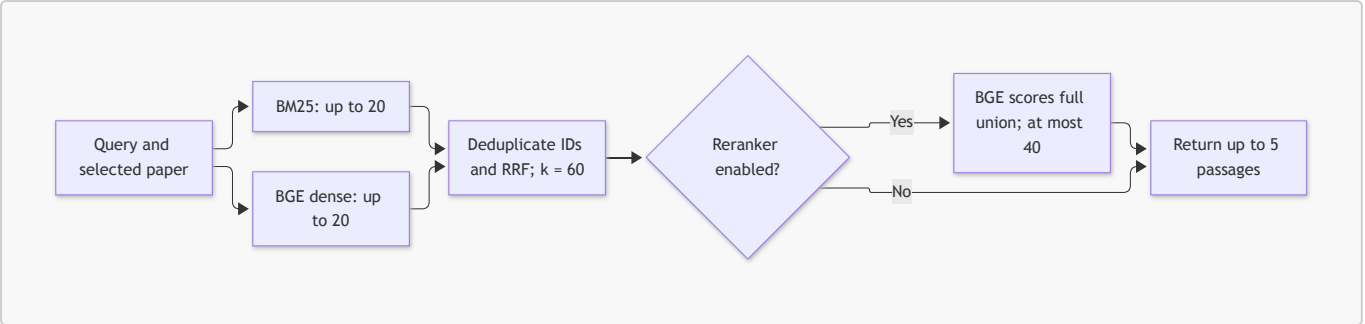
Explore some innovations for enhancing the retrieval and ranking. Explain why they are important to solve specific problems, illustrated with examples. You can list anything that has helped improving your system from the first version to the last one, plus queries that did not work earlier but now work because of the improvements you made. Because retrieval sets a ceiling on everything downstream, evaluate it explicitly with rank-aware measures such as Precision@K, Recall@K, Mean Reciprocal Rank, and Normalized Discounted Cumulative Gain (e.g., using ir measures, BEIR, or MTEB for embedding quality). Possible innovations include (but are not limited to) the following:

- Hybrid fusion (e.g., reciprocal rank fusion of sparse and dense results)
- Re-ranking (e.g., a cross-encoder or a learned re-ranker over the top candidates)

- Query expansion or rewriting (e.g., pseudo-relevance feedback, or HyDE-style hypothetical documents)
- Timeline and metadata search (e.g., allow users to search within specific time windows or categories)
- Multimodal search (e.g., implement image or table retrieval alongside text)
- Multilingual search (e.g., enable your system to retrieve data in multiple languages)
- Chunking and indexing strategies (e.g., semantic chunking or parent-document retrieval)

Field	Details
Status:	• In Progress — ★★★☆☆ (3/5)
Answer:	• BM25 + BGE dense retrieval → RRF → optional BGE cross-encoder reranking; section filtering and gap-directed rewriting are implemented. • Development chunk-size results and frozen four-method rankings exist; final relevance metrics await completed judgments. • See supporting details and any standalone tables below.
Decision / rationale:	• Compare saved native ranks on identical questions; treat pooled recall/nDCG as pool-relative. • 365 words is a documented tradeoff, not the primary-metric winner; no final ranking-quality gain is established.
Evidence / links:	• Hybrid implementation • Dense implementation • Pooled metrics implementation • Development chunk-size comparison • Frozen four-method settings • P2 historical handoff
Owner / updated:	• Responsible roles: P1/P2; P3 for chunking/labels; P4 for retries; P6 for metrics. • Named assignees/approval: pending confirmation. • Workbook entry updated: 2026-11-03; underlying evidence remains at the pinned snapshot.

Flow / workflow



- This diagram describes the main paper-scoped path; PR #20 adds a corpus-wide path with an optional paper filter.

Answer details

- The main hybrid pipeline combines complementary lexical and semantic evidence. BM25 uses casefolded Unicode word tokens (`k1=1.5`, `b=0.75`); dense retrieval uses `BAAI/bge-small-en-v1.5`, revision `5c38ec7c405ec4b44b94cc5a9bb96e735b38267a`, with L2-normalized embeddings and FAISS `IndexFlatIP` cosine search. The dense query instruction is "Represent this sentence for searching relevant passages: ". Lexical matching can preserve named methods or datasets; semantic similarity can retrieve paraphrases. These are design motivations, not measured improvements.

- Each source returns up to 20 candidates. Passage-ID deduplication and equal-weight reciprocal rank fusion use `RRF(p) = sum_s 1 / (60 + rank_s(p))`. This avoids adding incomparable BM25 and cosine scores. Optional `BAAI/bge-reranker-base` (revision `2cfc18c9415c912f9d8155881c133215df768a70`) scores the full unique union, at most 40 candidates, before selecting five. It uses query/passage pairs, raw logits, a 512-token limit and batch size eight in the frozen P3 configuration. Reranker ties use RRF score then passage ID. Short or empty pools remain short/empty; results are never padded.

- Additional mechanisms are exact-section filtering, score caching keyed by content/configuration, and P4 gap-directed query rewriting with section selection and whole-paper fallback. For a question about an experiment's metric, a missing-experiment-context diagnosis can target the paper's experiment section instead of repeating the original query. This is an illustrative mechanism, not an observed before/after success case. PR #20 additionally implements global BM25/dense rankings with optional paper filtering; the current formal evaluation remains paper-scoped.

- Chunking was selected using development-only BM25 experiments. The last round compared 325/365/375 words with fixed 40-word overlap, on 200 new questions plus 50 regression questions. On the new set, 365-word chunks achieved evidence-hit@5 **147/200** and complete-evidence@5 **125/200**, versus **145/200** and **123/200** for 375. However, 375 scored **108/200** versus **106/200** for 365 on the predeclared primary measure, complete evidence under a 1,000-word budget. Therefore 365 was a practical tradeoff, not the primary-metric winner or a statistically established optimum. Other overlap values were not compared. These development questions are excluded from final evaluation.

The four formal ranking arms already exist in P3's v3 pool: BM25, Dense, Hybrid RRF, and Hybrid RRF + BGE. Each has 1,000 top-five hits over 200 questions, with 1,853 unique pooled query–passage pairs.

Final metrics are pending completion of independent annotation, agreement calculation and adjudication:

Method	Precision@5	Pool-relative Recall@5	MRR@5	Pool-relative nDCG@5
BM25	Pending	Pending	Pending	Pending
Dense	Pending	Pending	Pending	Pending
Hybrid RRF	Pending	Pending	Pending	Pending
Hybrid RRF + BGE	Pending	Pending	Pending	Pending

- The historical 100-training-question P2 handoff changed the BGE top-five order/selection on all 97 nonempty questions and exactly reproduced P1's RRF order. This establishes implementation behavior, not better relevance. No verified final examples of “failed before, succeeds now” are available yet; add reviewed examples after scoring. Do not substitute synthetic metric tests or ranking-change counts for quality results.

Decision / rationale details

- Evaluate saved native ranks on the same questions; do not compress or re-sort each method into a different judged-only order without labeling that separate analysis. Precision divides by K; reciprocal rank uses the first relevant hit; binary nDCG discounts ranks logarithmically. Recall and ideal DCG derived from the pool are pool-relative, not exhaustive full-paper relevance. Unpooled passages are unknown, not confirmed negatives. Report coverage, exclusions and no-relevant-question handling, and bootstrap over questions/papers rather than correlated pairs.

3.3 Downstream Task and Generation (40 points)

Use the retrieved evidence to solve your chosen task. For a generative task (question answering, summarization, fact-checking), condition a language model on the retrieved passages, so that the answer is grounded in evidence a user can inspect rather than produced from the model’s parameters alone. For a classification task such as sentiment analysis, which is actually a complex suitcase research problem of many subtasks, you may feed the retrieved evidence to a language model as context or use it as features for a dedicated classifier; unless you are sure that your data does not contain neutral content, you should cover at least subjectivity detection and polarity detection (first neutral versus opinionated, then positive versus negative). Different approaches can be applied, including:

- knowledge based, e.g., SenticNet https://sentic.net
- rule based, e.g., linguistic patterns
- machine learning based, e.g., fine-tuned transformers or deep neural networks
- hybrid (a combination of any of the above, i.e., symbolic and subsymbolic AI)

You can tap into any resource or toolkit you like, as long as you motivate your choices and you are able to critically analyze obtained results. Some possible choices include:

- Hugging Face Transformers: https://github.com/huggingface/transformers

- PyTorch: https://pytorch.org
- TensorFlow / Keras: https://www.tensorflow.org, https://keras.io
- SciKit-learn: https://scikit-learn.org
- spaCy: https://spacy.io
- NLTK: https://www.nltk.org
- Local LLM serving: Ollama, vLLM, llama.cpp
- Open-weight model families (on the Hugging Face Hub): Llama, Mistral, Qwen, Gemma, Phi, DeepSeek
- Hosted LLM APIs (optional): OpenAI, Anthropic, Google, Cohere

Question 4

Perform the following tasks:

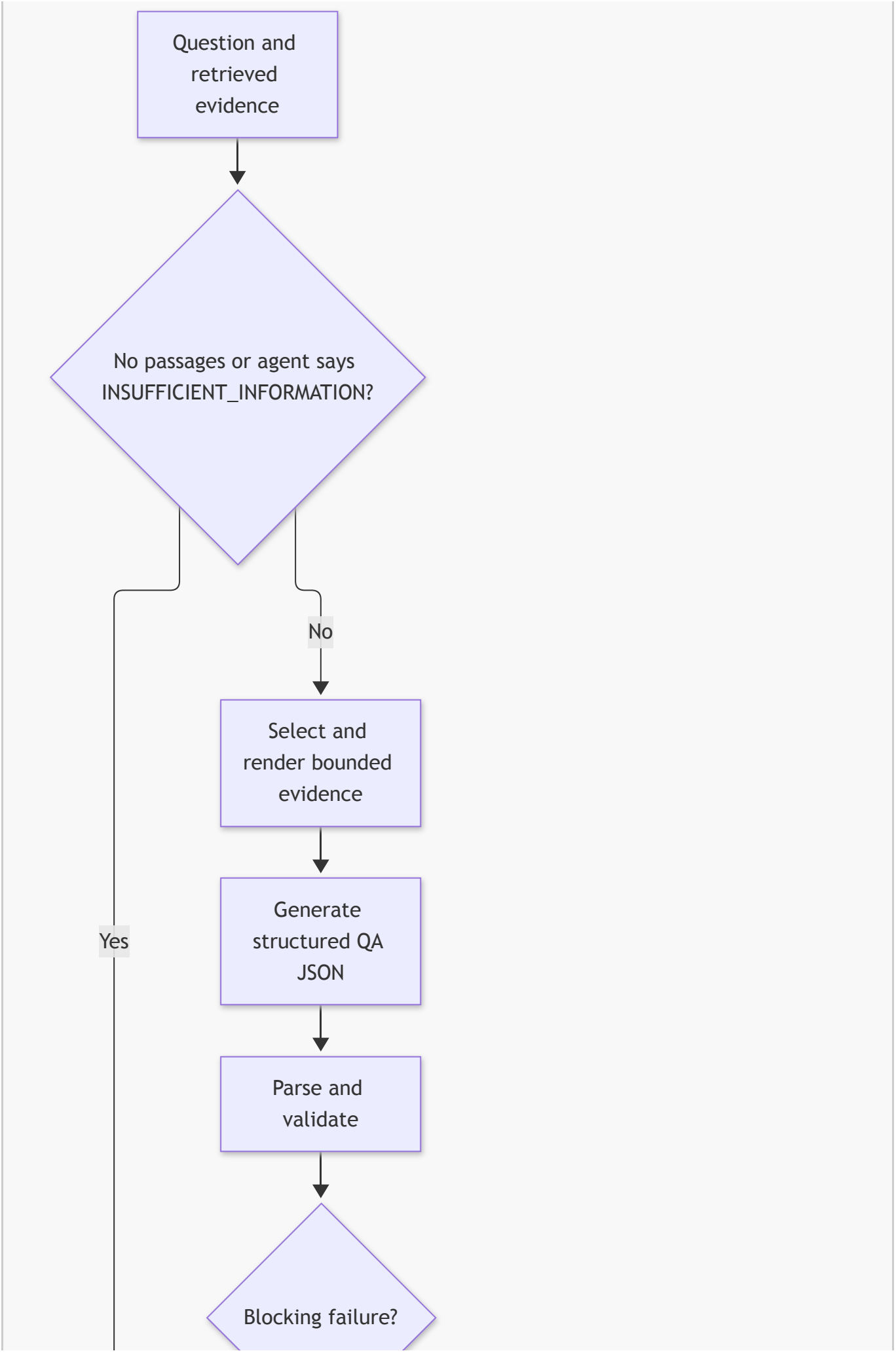
Q4.1

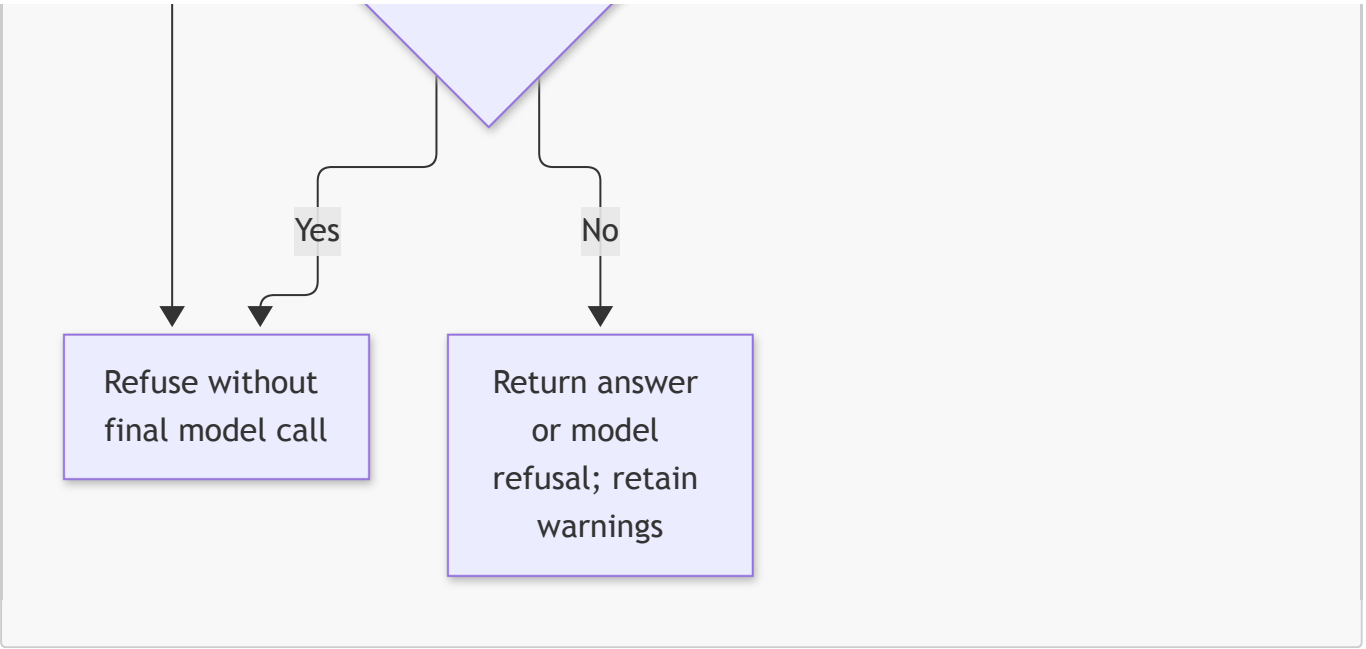
Motivate the choice of your generation or classification approach in relation with the state of the art

Field	Details
Status:	• Completed • This scoped response is supported by the inspected evidence; not a claim of project-wide completion.
Answer:	• Evidence-conditioned QA with structured JSON, citations, provenance, refusal and deterministic validation. • School path: Qwen3-8B; local PR path: Ollama Qwen3-14B. Context selection/truncation and validation limits are documented. • See supporting details and any standalone tables below.
Decision / rationale:	• Use generation for scientific answer synthesis, with inspectable supporting passages. • No external SOTA superiority claim; heuristic claim coverage and valid citation IDs do not prove entailment.

Field	Details
Evidence / links:	• Grounded prompt and context selection • Generation pipeline • Citation checks • School generation integration • Local backend and limits
Owner / updated:	• Responsible roles: P5; P4 for agent handoff; P6 for local integration. • Named assignees/approval: pending confirmation. • Workbook entry updated: 2026-11-03; underlying evidence remains at the pinned snapshot.

Flow / workflow


Question and retrieved evidence
No passages or agent says INSUFFICIENT_INFORMATION?
Yes
No
Select and render bounded evidence
Generate structured QA JSON
Parse and validate
Blocking failure?



- Valid citation membership and heuristic validation do not establish semantic correctness.

Answer details

- PaperGap uses an instruction-following language model conditioned on retrieved scientific passages. The school generation path uses **Qwen/Qwen3-8B**, with Transformers and vLLM backends; the PR's prepared Apple-machine development path instead uses Ollama **qwen3:14b**. These model configurations must not share performance claims. No project fine-tuning is established by the reviewed code.

- This design fits scientific QA because answers can require prose synthesis across passages, while citations let users inspect the evidence. The main prompt (`p5-grounded-qa-v7`) requires a structured answer with `answer`, `citations`, `evidence_status`, `retrieval_rounds` and `provenance`. It supplies a closed citation-ID list, prohibits unsupported numerical derivations and instructs refusal when evidence is missing. Qwen thinking is disabled for the JSON path. The documented school runner uses a 1,024-token output budget; backend and decoding settings must be taken from the actual run record rather than inferred from defaults.

- The pipeline selects/render-limits evidence, generates, parses and then validates. Empty evidence or the agent's `INSUFFICIENT_INFORMATION` label can refuse without a model call. A generic `sufficient=False` does not by itself suppress generation; it is retained as metadata. Main's prompt defaults allow at most eight passages and 1,200 characters per passage; top-five retrieval normally supplies fewer. Thus the generator may see only a prefix of a 365-word chunk. The prompt and validator share the rendered scope, and that truncation is a meaningful limitation for evidence coverage.

- Citation scope and numerical checks are deterministic, but claim coverage uses heuristics and can retain warning-level problems. Citation validity does not prove entailment or answer correctness. The approach follows the retrieval-conditioned generation pattern in the assignment; the repo does not contain a completed comparison against external state-of-the-art QA systems, so no SOTA performance claim is made. Local models offer control over prompts and validation, but their quality/cost advantage over hosted alternatives has not been established experimentally.

Q4.2

Discuss whether you had to preprocess data (e.g., microtext normalization) and why

Field	Details
Status:	• Completed • This scoped response is supported by the inspected evidence; not a claim of project-wide completion.
Answer:	• Normalize line endings/outer whitespace and chunk scientific full text without aggressive rewriting. • Apply BM25 tokenization, BGE query instructions, embedding normalization and model context limits separately. • See supporting details and any standalone tables below.
Decision / rationale:	• Preserve technical terms, numbers, equations and source placeholders. • Keep QA supervision out of indexed evidence; current annotation isolation is question-level, with shared papers allowed.
Evidence / links:	• Normalization/chunking • BM25 • Dense retrieval • Current question isolation policy • Generation parsing boundary
Owner / updated:	• Responsible roles: P3; P1/P2 for retrieval preprocessing; P5 for prompt/parse handling. • Named assignees/approval: pending confirmation. • Workbook entry updated: 2026-11-03; underlying evidence remains at the pinned snapshot.

Answer details

- QASPER contains scientific full text rather than informal social-media messages. We normalize line endings and outer whitespace, discard empty containers, and chunk within original section boundaries. We retain technical terms, case, numbers, equations, punctuation and source placeholders. Removing stopwords, lowercasing stored passages or aggressively rewriting text could damage the generator's evidence and numeric checks; those transformations are not applied to the stored scientific evidence.
- Retrieval preprocessing is separate: BM25 casefolds/tokenizes text, while BGE adds the retrieval instruction to queries and normalizes embeddings. The reranker tokenizes query–passage pairs with a 512-token limit. Generation selects passage IDs, renders text within its character budget and

requires structured JSON; parsing records format drift and can repair some invalid JSON escapes. These are model-input/output adaptations, not changes to the original gold labels.

Decision / rationale details

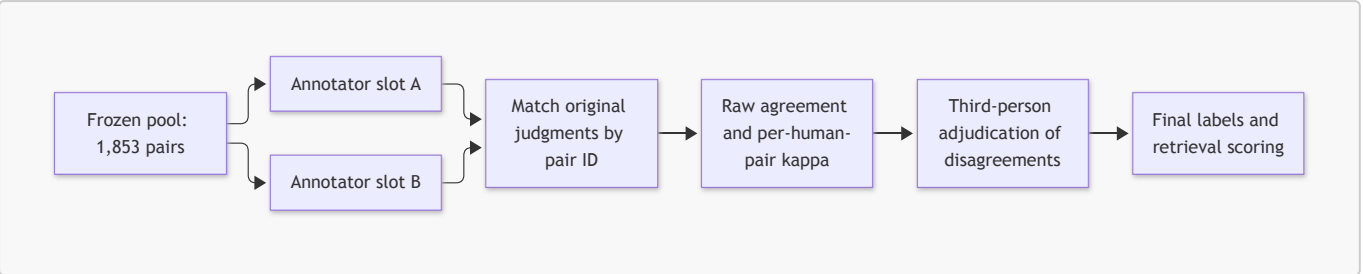
- Answers, evidence annotations and held-out relevance judgments are kept out of indexed passages and model query inputs. The formal annotation uses question-level exclusion of previously used questions; shared papers are allowed. This differs from the older proposed official-split-to-project-role mapping, which remains `v0-proposed`. Source split provenance must not be confused with a guarantee that the final question set is paper-disjoint.

Q4.3

Build an evaluation dataset by manually labeling at least 1,000 records with an inter-annotator agreement of at least 80% (it is recommended to have 3 annotators, but 2 is also OK)

Field	Details
Status:	• In Progress — ★★★★☆ (4/5)
Answer:	• 1,853 unique pairs; 3,706 planned judgments; 3,334 filled entries in the inspected snapshot. • B09/B10 contain 372 missing labels. Agreement, kappa, adjudication and final labeled export remain pending. • See supporting details and any standalone tables below.
Decision / rationale:	• Calculate agreement before adjudication and preserve independent labels; report actual human-pair assignments. • Disclose visible retrieval-method membership; do not claim fully method-blind annotation or ≥80% agreement yet.
Evidence / links:	• Current pool manifest • Batch sizes and output identities • B batch 09 • B batch 10 • Agreement protocol • Current usage boundary
Owner / updated:	• Responsible roles: All annotators; P3 for pool; P6 for agreement/adjudication coordination. • Named assignees/approval: pending confirmation. • Workbook entry updated: 2026-11-03; underlying evidence remains at the pinned snapshot.

Flow / workflow



- This is the intended completion sequence. B09/B10, agreement and adjudication are still pending; method membership is visible in the sheets.

Answer details

- The team recorded that unique query–passage pairs count as evaluation records. P3's current formal four-method pool contains **200 questions from 174 papers and 1,853 unique pairs**, exceeding the 1,000-record minimum once labeling is completed. Each method contributes 1,000 native top-five hits; a pair returned by several methods is labeled once per annotator, not counted repeatedly. The common pool has 6–14 distinct passages per question, so it is not the older proposed “100 questions × 10 passages” dataset.

- There are two annotator slots, A and B, each with ten batches of 20 complete questions, for **3,706 planned initial judgments**. Labels are binary: 0 = not relevant and 1 = relevant. In the fetched P3 snapshot, A's ten batches contain 1,853 filled labels; B's batches 01–08 contain 1,481, while **B/batch_09.md (177 pairs) and B/batch_10.md (195 pairs) are blank**. That is 3,334 filled entries and 372 missing entries. These counts verify file completeness only, not correctness, independent authorship or human review quality. They match the user's statement that two labeling files remain unfinished.

- The saved question list contains 78 official-validation and 122 official-test questions. P3 documents question-level isolation across official splits, with shared papers permitted, a scoped inventory of 2,654 excluded prior-use questions, and input/configuration hashes. That inventory does not establish coverage of unrecorded use by every team member. The pool is frozen for this retrieval comparison; this is not proof of a frozen generator/agent evaluation. The old paper-disjoint split mapping and passage schemas remain `v0-proposed` until confirmed.

Decision / rationale details

- **Protocol limitation:** the Markdown sheets display “Retrieved by” method names and grouped method membership. Although A/B are separate sheets, this is not fully method-blind annotation. Do not claim that retrieval-system identity was hidden. Document how annotators were assigned, whether they saw one another's labels, and the actual interpretation of partial evidence. Pair-level class balance is not established by the random question draw and must be reported after labels are finalized.

- After the remaining independent labels arrive, match labels by canonical question/passage identity, preserve both originals, and calculate raw agreement **before adjudication**. Report overall and per fixed human pair, with sample counts and Cohen's kappa per actual pair. A/B slot names alone do not prove that the same two humans annotated all batches. A third person can adjudicate

disagreements afterward; adjudication must not be counted as initial agreement. If agreement is below 80%, report it honestly and document guideline clarification/fresh independent labeling rather than overwriting disagreements.

- **Pending results:** final positive/negative counts, raw agreement, pair-specific kappa, disagreement counts, adjudicated labels and a verified final labeled export. No completed evaluation or ≥80% claim is made.

Q4.4

Provide task-appropriate quality metrics on such dataset: precision, recall, and F-measure for classification; Exact Match and F1 for extractive QA; ROUGE or BERTScore for summarization

Field	Details
Status:	• In Progress — ★☆☆☆☆ (1/5)
Answer:	• QA EM/token-F1 protocol is outlined; final generated-answer exports and scores are pending. • Retrieval relevance judgments do not replace question-level reference answers. • See supporting details and any standalone tables below.
Decision / rationale:	• Freeze normalization, multiple-reference, answer-type, abstention and coverage rules before scoring; pending confirmation. • Keep gold annotations outside model inputs and use matched questions/settings across arms.
Evidence / links:	• Evaluation protocol • QA export/access boundary • Historical generation observations
Owner / updated:	• Responsible roles: P5/P6; P3 for reference linkage. • Named assignees/approval: pending confirmation. • Workbook entry updated: 2026-11-03; underlying evidence remains at the pinned snapshot.

Answer details

- The task is scientific QA. Report normalized Exact Match and token F1 against eligible QASPER reference answers, with an explicit policy for multiple annotations, free-form/extractive/yes-no answers and abstentions. EM/F1 are interpretable baselines but exact wording can penalize valid paraphrases; report answer-type and answerable/unanswerable breakdowns and complement them

with grounding review. This workbook records the intended scoring policy, not a completed QA scorer/run.

Required QA result	Current status
Exact Match	Pending final generated-answer export and scoring protocol
Token F1	Pending final generated-answer export and scoring protocol
Answerable/unanswerable and answer-type breakdowns	Pending
Refusal/error coverage and multiple-reference policy	Must be fixed and recorded before final scoring

Decision / rationale details

- The manually labeled query–passage records evaluate retrieval relevance. They cannot by themselves establish QA Exact Match/F1 or replace question-level reference answers. Join generated answers to the corresponding saved question IDs, read gold/reference annotations only in the scoring stage, and never pass a full annotated QA row into the model. Use identical frozen questions, prompt, context budget and decoding across systems.
- P5's historical 100-training-question runs report answer/refusal and validation counts, not EM/F1. Those development observations must not be relabeled as held-out QA accuracy. The final evaluation is not done, as confirmed by the user.

Q4.5

Provide answer-level RAG metrics, i.e., faithfulness (is every claim grounded in the retrieved passages?), answer relevance, and context relevance/precision, e.g., using RAGAS

Field	Details
Status:	• Not started
Answer:	• Faithfulness, answer relevance and context relevance/precision are required; final RAGAS execution/results are unavailable. • The proposed judge backend is unset; package/metric definitions, judge settings and cost handling remain to be fixed. • See supporting details and any standalone tables below.

Field	Details
Decision / rationale:	• Proposed evaluation setup is pending confirmation; pin judge/embedding backends and refusal/missing-score policy before running. • Score actual rendered context; report deterministic validator diagnostics separately from RAGAS quality.
Evidence / links:	• Required evaluation and limitations • Unconsumed proposed evaluator settings • Validator implementation • Observed local warnings
Owner / updated:	• Responsible roles: P6; P5 for answer/context exports; judge/backend owner pending confirmation. • Named assignees/approval: pending confirmation. • Workbook entry updated: 2026-11-03; underlying evidence remains at the pinned snapshot.

Answer details

- Evaluate **faithfulness** (whether answer claims are supported by supplied passages), **answer relevance** (whether the answer addresses the query), and **context relevance/precision** (whether retrieved context is useful). These complement QA overlap metrics: a fluent answer may be wrong, a relevant context may not be used faithfully, and a valid citation ID may point to a passage that does not entail the claim.

Metric	Final value
Faithfulness	Pending
Answer relevance	Pending
Context relevance/precision	Pending

Decision / rationale details

- No completed, reproducible final RAGAS table was found in the reviewed sources. The main proposal still has `judge_model: null`. Before execution, pin the RAGAS package and exact metric APIs/definitions, judge model/revision, judge prompt/decoding, required embeddings, costs and treatment of refusals, empty contexts, failures and missing scores. Use the actual context shown to the generator when measuring answer faithfulness, recording any passage truncation. Keep retrieval-pool judgments and answer-level judge results separate.
- P5's deterministic validator checks citation membership and numbers and uses heuristic claim/provenance checks. Its warning counts, refusal counts or zero unknown-citation counts are diagnostics, not a substitute for faithfulness or human correctness evaluation. Local PR runs with `UNSUPPORTED_CLAIM` are explicit evidence that a returned answer is not automatically trustworthy.

Q4.6

Discuss performance metrics, e.g., latency, records processed per second, cost, and scalability of the system

Field	Details
Status:	• In Progress — ★★★☆☆ (3/5)
Answer:	• Historical warmed and non-warmed retrieval timings plus local UI observations are documented below. • Final matched latency, throughput, memory, token/cost and scalability measurements are pending. • See supporting details and any standalone tables below.
Decision / rationale:	• Separate historical corpus, current corpus, local Ollama, school vLLM, cold/warm/cache-hit and component/HTTP timings. • Record hardware and sample size; one browser duration is not throughput or a final benchmark.
Evidence / links:	• P2 documented timing results • Local integration observations • Pool runtime boundary • Server concurrency behavior
Owner / updated:	• Responsible roles: P2 for retrieval timings; P5 for generation; P6 for end-to-end measurements. • Named assignees/approval: pending confirmation. • Workbook entry updated: 2026-11-03; underlying evidence remains at the pinned snapshot.

Answer details

The historical P2 six-training-question warmed pilot on macOS ARM64 CPU reported:

Candidates per source	RRF p50 / p95 (ms)	RRF + BGE p50 / p95 (ms)
10	10.1 / 10.4	1,835.3 / 2,147.8
20	10.3 / 11.9	3,172.5 / 4,151.5

- Both paths were warmed; model loading and index construction were excluded. The index was built on Metal, but retrieval/reranking ran on CPU with four threads. These are small historical development measurements, not the current 365/40 corpus or end-to-end UI benchmark. They suggest reranking dominates that CPU pipeline; they do not establish final quality or throughput.

- A separate historical 100-question integration run, without warm-up, reported RRF p50/p95 **17.57/21.34 ms** and RRF+BGE **2,387.34/3,768.72 ms**. It included three empty retrievals per arm and first-call costs (5,904.84 and 9,280.55 ms). Summed retrieval-call times, 7.61 and 243.92 seconds, exclude setup outside calls and are not process wall time. These figures are documented by P2, not remeasured during this edit.

- The local PR UI runs discussed in Q2.2 took approximately 75–79 seconds per observed request on Apple M5/24 GB with Ollama Qwen3-14B and adaptive retrieval. Those durations include HTTP waiting, retrieval, critic and final-answer work and must not be mixed with P2 component timings or school Qwen3-8B performance. P3's 2,128.66-second formal pool-building runtime includes indexing and all four methods and is explicitly **not a latency benchmark**.

- Scalability/cost considerations: BM25 statistics and FAISS flat vectors are built/loaded before queries; flat dense search scales with the indexed vectors, and reranking is limited to the source top-20 union. Up to two additional agent retrieval rounds add retrieval/reranker and critic work. Persistent reranker cache hits can make repeated timing unrepresentative. The P5 server serializes answer work and returns HTTP 429 when busy; this is not a measured concurrent production service. Local Ollama requires no hosted per-call fee, but hardware, energy and runtime are not free. School GPU availability/quota and judge costs still affect feasibility.

Decision / rationale details

- **Pending:** matched final p50/p95 end-to-end and component latency, records/second, token/cost accounting, memory usage and load/scalability measurements. Fix hardware/model/context/threads, record cold versus warmed versus cached runs, preserve errors/refusals, and report sample size before claiming performance. Do not calculate “throughput” as the inverse of one browser request.

Question 5

Explore some innovations for enhancing the downstream task. If you introduce more than one, perform an ablation study to show the contribution of each innovation. For example, if you add re-ranking and query rewriting, show the gain from adding only re-ranking, the gain from adding only query rewriting, and the gain from adding both. Explain why they are important to solve specific problems, illustrated with examples. Possible innovations include (but are not limited to) the following:

- Better grounding (e.g., cite sources, or add a faithfulness/self-check step that flags unsupported claims)

- Advanced prompting or reasoning (e.g., chain-of-thought, self-consistency)

- Iterative or agentic RAG (e.g., let the system issue follow-up retrievals until it has enough evidence)

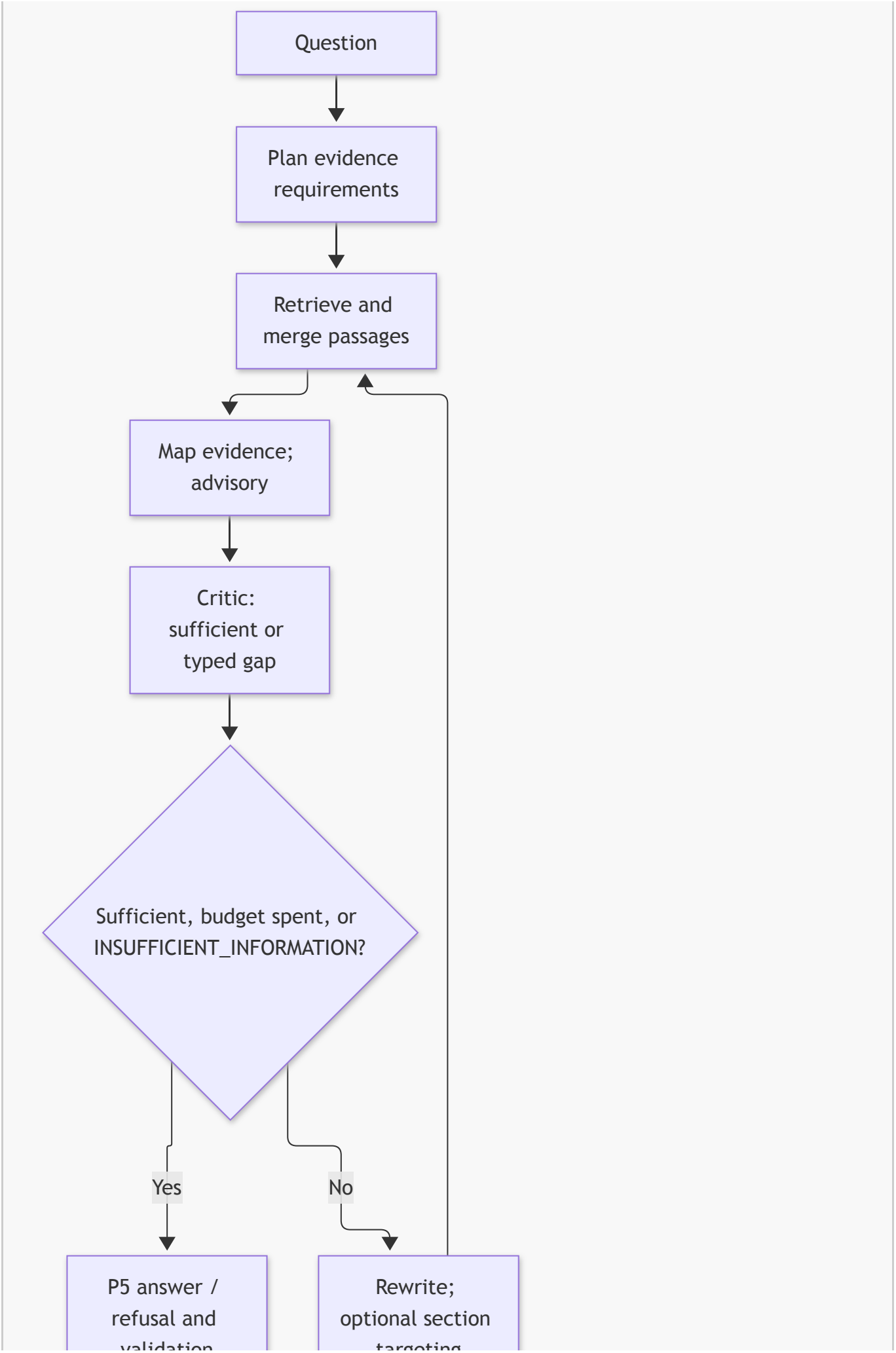
- Fine-tuning or instruction-tuning the generator on your task

- Enhanced classification (e.g., add a subtask such as sarcasm detection, or perform two tasks jointly)

- Fine-grained classification (e.g., perform aspect-based sentiment analysis)

Field	Details
Status:	• In Progress — ★★★☆☆ (3/5)
Answer:	• Gap-directed agent retries and evidence-constrained generation/validation are implemented; development diagnostics exist. • Final controlled ablations and verified before/after quality examples are pending. • See supporting details and any standalone tables below.
Decision / rationale:	• Use matched generator/prompt/context/questions and test reranking-only, rewriting-only, both and neither when claiming separate contributions. • Freeze generator/agent settings separately from the retrieval pool; no final adaptive superiority or hallucination-prevention guarantee.
Evidence / links:	• P4 flow and limitations • Adaptive demo factory and flags • Agent experimental measurements • P5 historical comparison • Grounding/validation pipeline • Local adaptive integration
Owner / updated:	• Responsible roles: P4/P5; P1/P2 for retrieval arms; P6 for controlled evaluation. • Named assignees/approval: pending confirmation. • Workbook entry updated: 2026-11-03; underlying evidence remains at the pinned snapshot.

Flow / workflow


Question
Plan evidence requirements
Retrieve and merge passages
Map evidence; advisory
Critic: sufficient or typed gap
Sufficient, budget spent, or INSUFFICIENT_INFORMATION?
Yes
No
P5 answer / refusal and
Rewrite; optional section

validation targeting

- At most two retries means three retrieval rounds total. The diagram describes implemented control flow, not a measured recovery benefit.

Answer details

- **Gap-directed adaptive retrieval.** P4 uses a LangGraph flow: plan → retrieve → map evidence → critic → either stop or rewrite and retrieve again. Gap labels distinguish missing method, experiment context, result, comparison, terminology mismatch, weak evidence and insufficient information. The rewriter incorporates paper terminology and missing components; section selection uses the paper's own headings and can fall back to whole-paper retrieval. The hard budget is two retries, at most three rounds. A result question lacking experimental context can trigger a targeted follow-up query rather than accepting the first top-five set. This example explains intended behavior, not a proven successful recovery rate.

- **Evidence-constrained generation and validation.** P5's JSON prompt supplies valid citation IDs, requires support for factual claims and numbers, and allows refusal. The pipeline validates the rendered evidence scope, flags heuristic claim-coverage/provenance problems, repairs some citation/format defects and downgrades specified error cases to refusal. This makes unsupported output inspectable and sometimes blocks it; it does not guarantee semantic faithfulness. Deterministically backfilled provenance is a diagnostic map, not independent evidence of entailment.

- **Retrieval-to-answer integration.** RRF and optional BGE reranking choose the evidence the generator sees. Historical development results show why “more sophisticated” is not automatically better: P5's comparable v7 100-training-question RRF baseline answered 85 and refused 15; BGE top-five answered 79 and refused 21. The reranked arm had nine answered→refused and three refused→answered flips. Six of the nine lost every passage cited by the old answer. These are development diagnostics on the older corpus, not gold-scored accuracy, and they do not establish a causal quality gain. The reranked rerun recorded zero error-level validator violations but still 21 `PROVENANCE_OVERSTATED` warnings; fewer errors is not proof of better answers.

- The main `scripts/demo_agent_factory.py` provides fresh graph execution via the P5 record-factory hook, reusing the generator for criticism. Reranking defaults to **off** in that factory unless enabled; the PR's prepared local path explicitly enables BGE and adds corpus-wide retrieval. Thus report actual runtime flags, backend and scope, not a single universal configuration. Rule-based evidence mapping is advisory; Qwen critic performance on the school real-model evaluation remains unvalidated. The PR documents local Ollama critic execution, including warnings/refusal cases, which is a separate integration result.

Controlled ablation plan - not completed results.

Comparison	Hold fixed	Contribution tested	Final result
BM25 / Dense / RRF / RRF+BGE	Questions, corpus, candidate depth, scoring protocol	Lexical/semantic combination and reranking	Pending independent final labels

Comparison	Hold fixed	Contribution tested	Final result
Static RRF vs static RRF+BGE QA	Generator, prompt, context budget, validation, questions	Reranking's downstream effect	Historical development diagnostics only; final QA/RAGAS pending
Static RRF+BGE vs adaptive RRF+BGE QA	Same answer generator/prompt, validation, context budget and questions	Critic/rewriting/retries as a combined module	Pending final controlled execution
Rewriting only / reranking only / both / neither	Same generator, retrieval scope, questions and other settings	Their individual and joint effects	Pending; do not infer a factorial result from the previous comparisons
Validator on vs off, if performed	Same raw generated answers and retrieval inputs	Deterministic postprocessing/abstention tradeoff	Optional, not measured here

Decision / rationale details

- For each completed arm report QA EM/F1, RAGAS, citations/refusals/errors, retrieval recall where valid gold mappings exist, round counts, critic fallback, recovery/ineffective-retry rates and latency/token cost. Use matched question-level differences and uncertainty. Freeze generator/agent prompts and settings separately from P3's already frozen retrieval pool; do not tune from held-out labels or reuse selected chunk-size questions as final test data. Add reviewed before/after examples only after outputs and judgments exist.
- **Decision:** no final ablation gain, adaptive superiority or universal hallucination-prevention claim can currently be made. Fine-tuning, multilingual/multimodal retrieval, dual critics and NLI entailment checking are not established delivered innovations in this answer.

4 Submission

Select one person of the group in charge of submitting the final assignment report. Submission has to be done via Blackboard (do not email your report). As you write your report, it is strongly suggested that you also prepare some slides along with it because your work will also have to be presented in person on week 13. The submission shall consist of one single PDF file named after your group number, e.g., if you are group 10, your file should be titled simply “10.pdf”. Failing to name the file correctly or sending it in the wrong format, e.g., zip or MS Word, will result in demerit points. Do add some pictures to your report to make it clearer and easier to read. There is no page limit and no special formatting is required. The file shall contain the following five key items:

1. The names of all group members in the first page
2. The matriculation number of all group members in the first page

3. Your answers to all the above questions
4. A Dropbox (or similar, e.g., Google Drive or OneDrive) link to a compressed (e.g., zip) file with your knowledge base, queries and their retrieved results, evaluation dataset, generated answers or classification results, and any other data for Questions 3 and 5
5. A Dropbox (or similar, e.g., Google Drive or OneDrive) link to a compressed (e.g., zip) file with all your source codes and libraries, with a README file that explains how to compile and run the source codes

Submission preparation status (added)

All five questions now have repository-backed draft answers, including explicit gaps. Before the final PDF, add the group number, all member names and matriculation numbers, the nominated submitter, report figures, and verified code/data/result archive links with access permissions checked. Final labels/agreement, five matched live-query exports, QA/RAGAS results and controlled ablation tables remain incomplete. Existing source/download links are not automatically the final submission archives. Name the final PDF after the actual group number as the assignment specifies; do not invent missing member details or archive URLs.